

Shardul Marathe

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EDUCATION

Stanford University

B.S in Computer Science, Minor in Economics

Stanford, CA

Expected 2029

- **GPA:** 3.9
- **Relevant Coursework:** Linear Algebra & Differential Equations (Fourier Methods); Programming Abstractions (DSA); Computer Organization & Systems; Strategic Economics & Data-Driven Decision Making; Electricity & Magnetism

RESEARCH EXPERIENCE

AlgoVerse AI Research Program – Applied LLM Evaluation Research

Remote

Machine Learning Researcher (Mentored by Sunishchal Dev)

Feb 2025 - July 2025

- Designed and ran large-scale experiments testing whether autonomous AI coding agents would follow hidden malicious instructions while completing legitimate tasks, using environments in Docker and distributed computing on RunPod
- Built adversarial task suites in Python that introduced covert side objectives such as injected logic errors and unsafe numerical optimizations, systematically measuring failure rates across model variants and prompting strategies
- Engineered logging and analysis pipelines using Python, Pandas, and structured JSON outputs to capture execution traces, tool calls, and code diffs, enabling quantitative analysis of deceptive behavior patterns
- Developed and benchmarked a monitoring framework using separate “trusted” models to detect suspicious activity, comparing detection performance across prompting strategies and contributing to a NeurIPS 2025 workshop paper

Aspiring Scholars Directed Research Program – Computational Biology Research

Fremont, CA

Researcher (Supervised by Sahar Jahanikia)

Aug 2022 - Mar 2023

- Analyzed gene expression datasets from multiple neurodegenerative diseases to investigate shared structural patterns across disease cohorts
- Applied dimensionality reduction techniques (PCA, UMAP, t-SNE) to visualize similarities between diseases and explore latent structure in high-dimensional transcriptomic data
- Constructed and examined correlation matrices to identify recurring expression patterns and subgroup structure within and across diseases, with attention to noise and visualization artifacts
- Prepared quantitative visual analyses and research summaries for accepted presentation at the Organization for Human Brain Mapping (OHBM) Annual Conference

PROFESSIONAL EXPERIENCE

SpeechBoost AI - Speech Analysis Platform

Remote

Full Stack Developer

Aug 2023 - Present

- Built and deployed a full-stack web platform leveraging OpenAI’s LLM APIs to analyze humor, clarity, and emotional impact in user-submitted speeches, serving 2,000+ monthly active users
- Developed structured prompt templates and post-processing pipelines to convert raw model outputs into consistent scoring categories and actionable feedback
- Iterated on scoring rules and evaluation thresholds to improve feedback stability and reduce ambiguous outputs, securing \$50K+ in angel investment to expand multilingual capabilities

OwnABot Academy - Robotics Education Business

Fremont, CA

Owner & Web Developer

Jun 2021 - May 2025

- Scaled the program to 500+ enrolled students and \$102,000+ in cumulative revenue over four years
- Built a customer-facing platform to support large enrollments, robotics kit distribution, and parent communications
- Secured 3 physical locations across the San Francisco Bay Area to expand in-person instruction capacity

Robots Without Borders - International Robotics Education Nonprofit

Remote

Co-Founder & Web Developer

May 2023 - May 2025

- Founded an international STEM nonprofit delivering robotics programs across Africa, Asia, and North America, coordinating curriculum and instructors across time zones
- Designed and deployed solar-powered water purification buggies for underserved communities, applying robotics to address local clean water challenges
- Raised \$4,000+ through sponsorships and community fundraising to support equipment, logistics, and program expansion

TECHNICAL SKILLS

- **Languages:** Python (primary), Java, C++, C
- **ML & Data:** PyTorch, scikit-learn, Pandas, NumPy
- **Backend & Tools:** PostgreSQL, Supabase, Docker, Git, REST APIs, Vercel